

Dr. Gokay Avci

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TECHNICAL SKILLS

- **Programming:** Python, Rust
- **Scientific Computing & Modelling:** CP2K, Gaussian, RASPA, Zeo++, Materials Studio, CCDC suite, GULP
- **Data & AI:** PyTorch, high-throughput screening pipelines, SQLite, API-driven workflows
- **Workflow & Orchestration:** HPC job orchestration, bespoke automation scripts, database backed experiment tracking

EXPERIENCE

University College London, Department of Chemistry

Oct 2025

Research Fellow — Computational Chemistry and Physics

London, United Kingdom

- Supervisor: Prof. Scott Woodley
- Projects in progress within MCC.

Imperial College London, Department of Chemistry

Oct 2022 — Sep 2025

Research Associate — Computational Supramolecular Materials

London, United Kingdom

- Supervisor: Prof. Kim E. Jelfs.
- Discovered organic host–guest pairs using evolutionary algorithms, reducing computational cost by an order of magnitude versus brute-force search.
- Enabled synergistic computational/experimental discovery of high-symmetry porous organic cages from low-symmetry building blocks via high-throughput screening.
- Developed, tested, and maintained computational tools for materials discovery using quantum chemistry and atomistic simulation methods.

Koç University, Materials Science & Engineering

Sep 2016 — Dec 2021

Graduate Research Assistant (PhD Researcher)

Istanbul, Turkey

- Thesis: **High-Throughput Computational Screening of MOFs for Carbon Dioxide Capture and Hydrogen Purification.**
- Supervisor: Prof. Seda Keskin Avci.
- Performed atomistically detailed simulations on 10,000+ MOFs; identified next-generation adsorbents and membranes for CO₂/H₂ separations.
- Conducted extensive structure–property analysis using multimodal datasets derived from Grand Canonical Monte Carlo and Molecular Dynamics simulations.

Sabancı University, Materials Science & Engineering

Sep 2013 — Aug 2015

Graduate Research Assistant (MSc Researcher)

Istanbul, Turkey

- Thesis: **Investigation of machinability of green bodies of Solid Cast Alumina through the Addition of a Poly (Carboxylate Ether)-based Superplasticizer.**
- Supervisors: Asst. Prof. Özge Akbulut; Prof. Yusuf Z. Menceloğlu.
- Formulated machinable ceramic green bodies via addition of PCE-based rheology modifiers; outcomes contributed to a European patent and a first author paper.

EDUCATION

Koç University

Istanbul, Turkey

PhD, Materials Science & Engineering (2:1)

2016 — 2021

Sabancı University

Istanbul, Turkey

MSc, Materials Science & Engineering (2:1)

2013 — 2015

Sabancı University

Istanbul, Turkey

BSc, Materials Science & Engineering (2:1) — Minor: Chemistry

2009 — 2013

Üsküdar American Academy

Istanbul, Turkey

Math & Natural Sciences Track

2004 — 2009

AWARDS

- TYC Scientific Image Competition (2024) — **Best scientific image**.
- Sabancı University — Full merit scholarship.
- Koç University — Full merit scholarship.

PUBLICATIONS & PATENTS

- Lu W., Yao Y., Un H., **Avci G.**, Humid Z., Ai K., Tian J., Zhao L., Durrant R. J., Sirringhaus H., Jelfs E. K., McCulloch I. **Covalent Organic Framework as Organic Electrochemical Transistor Active Layer: Hydrophilicity, Porosity and Planarity** (in preparation).
- **Avci G.**, Turcani L., Jelfs E. K. **Computational Discovery of Novel Host/Guest Pairs with Evolutionary Algorithms** (in preparation).
- **Avci G.**, Hafizi R., Wolpert E. H., Humphreys J., Rabbani M., Basford A., Greenaway B., Cooper A. I., Day M. D., Jelfs E. K. **Discovery of High-Symmetry Porous Organic Cages Composed of Low-Symmetry Precursors with Multi-scale Computational Modelling** (in preparation).
- Yusaf Z., Egleston B., **Avci G.**, Jelfs E. K., Lewis J., Greenaway B. **Organic Cage Rotaxanes** (in revision).
- Lu W., Tait C., **Avci G.**, Li X., Shao P., Aitchison C., Ceugniet F., Yao Y., Frogley M., Decarolis D., Yao N., Jelfs E. K., McCulloch I. **Co-Embedded Metal-Covalent Organic Frameworks for CO₂ Photoreduction** *J. Am. Chem. Soc.* (2025).
- Markiewicz G., Qiu X., **Avci G.**, et al. **Chirality-Assisted Self-Assembly of Low-Symmetry Non-Covalent Capsules with Quantitative Diastereoisomeric Selection** (submitted / in revision).
- Brand M., Trowell H., Pegg J., Greenfield J., Odaybat M., Little M., Haycock P., **Avci G.**, Rankin N., Fuchter M., Jelfs E. K., Cooper A. I., Greenaway R. L. **Photoresponsive Organic Cages — Computationally Inspired Discovery of Azobenzene-Derived Organic Cages** *J. Am. Chem. Soc.* (2024).
- Zhu Q., Qu H., **Avci G.**, et al. **Computational Guided Synthesis of a Hierarchical [4[2+3] + 6] Porous Organic ‘Cage of Cages’ Nature Synthesis** (2024).
- **Avci G.**, Jelfs E. K. **Host–Guest Binders Nature Computational Science / Nature Computational Materials** (2024).
- **Avci G.**, Altintas C., Keskin S. **Metal Exchange Boosts the CO₂ Selectivity of MOFs Having Zn-oxide Node** *J. Phys. Chem. C* (2021).
- Daglar H., Gulbalkan H. C., **Avci G.**, et al. **Effect of MOF Database Selection on the Assessment of Gas Storage and Separation Potentials of MOFs** *Angew. Chem.* (2021).
- **Avci G.**, Erucar I., Keskin S. **Do New MOFs Perform Better for CO₂ Capture and H₂ Purification? Computational Screening of the Updated MOF Database** *ACS Appl. Mater. Interfaces* (2020).
- **Avci G.**, Velioglu S., Keskin S. **In-silico Design of MOFs with Enhanced CO₂ Separation Performances: Role of Metal Sites** *J. Phys. Chem. C* (2019).
- Altintas C., **Avci G.**, Daglar H., Azar A., Erucar I., Velioglu S., Keskin S. **Comparative Analysis of Two MOF Databases: High-Throughput Screening of Computation-Ready MOFs for CH₄ and H₂ Adsorption** *J. Mater. Chem. A* (2019).
- **Avci G.**, Velioglu S., Keskin S. **High-throughput Screening of MOF Adsorbents and Membranes for H₂ Purification and CO₂ Capture** *ACS Appl. Mater. Interfaces* (2018).
- Altintas C., **Avci G.**, Daglar H., Azar A., Velioglu S., Erucar I., Keskin S. **Database for CO₂ Separation Performances of MOFs Based on Computational Materials Screening** *ACS Appl. Mater. Interfaces* (2018).
- Altintas C., **Avci G.**, Daglar H., Gulcay E., Erucar I., Keskin S. **Computer Simulations of 4,240 MOF Membranes for H₂/CH₄ Separations** *J. Mater. Chem. A* (2018).
- **Avci G.**, Akgulaghi O., Ustbas B., Ozbay C., Akbulut O., Menciloglu Y. Z. **A PCE-based Rheology Modifier Allows Machining of Solid Cast Green Bodies of Alumina Ceramics** *International* (2015).
- European Patent 14178231.8–1958 A — **Composite Structure for Medical Training and Production Method Thereof** (Inventors: Özge Akbulut Halatci, Barkin Eldem, Ece Polen Budak, Gokay Avci, Ceren Ozbay).

REFERENCES

Available upon request.